
BudgetFlow: Value-Aware Budget Governance for Agent Tasks

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Abstract

Multi-step AI agent tasks consume variable amounts of model budget, and tasks in a batch often have unequal value. A strong model can define a good frontier point when broad capability pays for its cost. A learned router can define a good frontier point when per-task Model Fit is accurately predicted. We study a third view: *Batch-Level Task Budget Allocation*, where a task batch shares one hard budget and scarce model opportunities should flow toward the tasks that create the most verified value.

We introduce BudgetFlow, a value-aware budget governance framework for task batches. Its general interface combines task-level signals—Task Value, Estimated Token Cost, Model Fit, and a Verifier—with batch-level Shared Budget state. BudgetFlow uses the first three task signals and the Shared Budget to allocate budget and model tiers, while the Verifier settles outcomes. This separates allocation signals from outcome settlement, so the same interface can describe coding, support, document, spreadsheet, and research task batches once each domain supplies a trusted verifier.

We evaluate BudgetFlow on SWE-bench-style verified bug-fixing tasks under fixed task sets and shared hard budgets. The paper reports Resolved Rate, Total Resolved Value (TRV), and TRV per Dollar. Across one 4-policy run and three 5-policy runs over 30-task batches, BudgetFlow produces the latest audited run’s best Resolved Rate, TRV, and TRV per Dollar; it also exposes a value-efficiency gain in an earlier run where raw resolved count favors the strong-model-only baseline. The evidence is reported as frontier modes—Value Divergence, Strong Coverage, and Model Fit—showing when value-aware batch governance matters and how strong baselines form meaningful frontier boundaries.

1. Introduction

AI agents increasingly execute multi-step tasks: they inspect context, call tools, edit artifacts, run checks, and decide whether another model call is worth the cost. These tasks consume budget unevenly. One task may finish after a short reasoning loop; another may require repeated search, repair,

and verification. In a batch, this variation means that a fixed model budget can be exhausted before every task receives the same opportunity to succeed.

A central issue is that tasks rarely have equal value. A support ticket with an imminent service-level agreement, a security patch, a revenue-critical spreadsheet fix, and a routine rewrite should receive different budget treatment when they are submitted in the same batch. Once model calls become a scarce shared resource, the system needs an allocation policy in addition to a model router.

This paper studies *Batch-Level Task Budget Allocation*. The general setting has many tasks, one shared hard budget, model tiers with different costs and capabilities, and an outcome rule that decides whether a task is resolved. The objective is to allocate scarce model opportunities so the batch produces more verified task value under the same resource constraint.

The operational decision is simple: which task should receive the next scarce model opportunity, under the same Shared Budget?

Two existing operating views are already useful in this setting. The strong-model view spends on the strongest tier when its broad capability is worth the cost. The learned-routing view predicts which model tier fits each task. BudgetFlow adds the value-governance view: when tasks differ in value, estimated token cost, and model-tier upside, the batch policy should allocate scarce model opportunities across the full task batch.

BudgetFlow is a budget governance framework for this setting. It uses four task-level signals and one batch-level budget state:

- **Task Value:** the pre-registered value gained if the task is resolved;
- **Estimated Token Cost:** the pre-run estimate of token cost used for cap planning;
- **Model Fit:** evidence about how well each model tier fits the task;

- **Verifier:** the outcome rule that maps a completed attempt to resolved or unresolved;
- **Shared Budget:** the hard batch-level resource constraint shared by all tasks.

The first three task-level signals and the Shared Budget drive allocation decisions. The Verifier settles outcomes and defines the objective. This separation is the core interface. The policy can change how it routes, caps, assigns model opportunities, or stops tasks, while the outcome contract remains explicit. Task-source labels such as coding, marketing, support, or document production can be recorded as context; strong-model access is governed by declared value, estimated token cost, model-fit evidence, and the shared budget.

We instantiate this formulation on SWE-bench-style verified coding tasks. SWE-bench is useful here because it gives a strict task-batch testbed: each task is a real bug-fixing issue, each attempt has measurable model spend, and each outcome is settled by a test-based harness. The benchmark is the evidence domain, while the allocation interface is the research object.

The paper uses three primary metrics. **Resolved Rate** anchors the evaluation to standard verified success. **Total Resolved Value** (TRV) sums pre-registered Task Value over resolved tasks. **TRV per Dollar** measures value efficiency under the shared hard budget.

The central empirical result is the BudgetFlow frontier view. BudgetFlow wins the latest audited 5-policy, 30-task run on Resolved Rate, TRV, and TRV per Dollar. It also wins a 4-policy run on value and value efficiency while resolving one fewer task than the strong-model-only baseline. Strong-model-led and learned-router-led diagnostics are informative within the same framework: they mark operating conditions where broad strong-model coverage or accurate model-fit prediction forms a strong frontier point. Together, these outcomes show how the frontier is shaped by Shared Budget pressure, Task Value, Estimated Token Cost, and Model Fit.

This paper makes three contributions:

1. We formulate Batch-Level Task Budget Allocation as a value-governance problem with pre-registered Task Value, Estimated Token Cost, Model Fit, a shared hard budget, and a Verifier-settled objective.
2. We present a generalizable BudgetFlow interface that allocates model opportunities through Task Value, Estimated Token Cost, Model Fit, Shared Budget, and a Verifier-settled outcome contract across task domains.
3. We report frontier-mode evidence against cheap-model-only, strong-model-only, learned task-router, and

budget-only policies, identifying Value Divergence, Strong Coverage, and Model Fit as the operating views that explain when BudgetFlow moves the frontier and when strong controls define a boundary.

2. Related Work

Per-query routing and cascading. RouteLLM learns per-query routers from preference data (Ong et al., 2025). Cascade Routing studies model routing and cascading for individual queries (Dekoninck et al., 2025). RouteNLP targets enterprise query routing under quality constraints (Guo et al., 2026). UCCI calibrates uncertainty for cost-optimal cascade routing (Kotte, 2026). These works make model selection a first-class cost-quality problem. The unit is typically one prompt or one request. BudgetFlow operates at the task-batch layer: many multi-step tasks share one hard budget, and the policy allocates scarce model opportunities across tasks with different values.

Within-task budgeted agents. INTENT studies budget-constrained tool use inside a single agent task (Liu et al., 2026). This is a complementary layer to Batch-Level Task Budget Allocation. A within-task policy decides whether one task should spend another tool or model call. BudgetFlow decides how one budget should be shared across many tasks before the batch budget is exhausted.

Coding-agent evaluation and cost-aware runtimes. SWE-bench provides the verified coding-task substrate used by many agent evaluations (Jimenez et al., 2024). OpenSquilla is a token-efficient agent runtime with model routing, diagnostics, replay, and cost-per-success framing (OpenSquilla contributors, 2026). Claw-SWE-Bench studies harnesses, adapters, prompts, patch extraction, leakage cleanup, and cost reporting for SWE-bench-style coding-agent evaluation (Zheng et al., 2026). These systems sharpen the evaluation requirements for agent tasks: cost, harness behavior, and outcome verification must be explicit.

Serving substrates. vLLM improves LLM serving through PagedAttention and KV-cache management (Kwon et al., 2023); SGLang accelerates structured language-model programs with runtime optimizations such as KV-cache reuse (Zheng et al., 2024); Dynamo exposes agentic inference concerns including cache locality and scheduling (NVIDIA, 2026). These systems provide execution substrates. BudgetFlow provides task-level value and budget signals that can inform priority, continuation, model opportunity, and cache-sticky scheduling in future serving-aware systems.

3. Problem Formulation

Let a task batch contain tasks $\mathcal{T} = \{1, \dots, n\}$ and let B denote the Shared Budget. Each task i has a pre-registered value $v_i > 0$, an estimated token cost $d_i > 0$, and model-fit evidence $m_{i,k}$ for each model tier k . A policy π chooses a sequence of actions that spend budget on model calls and task attempts. Let $c_i(\pi)$ be the spend on task i , and let

$$\sum_{i=1}^n c_i(\pi) \leq B$$

be the shared hard-budget constraint.

The budget is shared at the batch level. A policy can spend more on one task only by leaving less budget for later tasks in the same batch. This coupling is what separates the problem from independent per-task routing: a locally reasonable strong-model call can be globally poor if it consumes budget needed by a higher-value task.

Each task has a Verifier V_i that maps the final task evidence to a binary outcome:

$$r_i(\pi) = V_i(\pi, i) \in \{0, 1\}.$$

The primary value objective is Total Resolved Value:

$$\text{TRV}(\pi) = \sum_{i=1}^n v_i r_i(\pi).$$

We report three headline metrics:

$$\text{ResolvedRate}(\pi) = \frac{1}{n} \sum_{i=1}^n r_i(\pi),$$

$$\text{TRVperDollar}(\pi) = \frac{\text{TRV}(\pi)}{\sum_i c_i(\pi)}.$$

3.1. Core Terms and Metric Reading

Table 1 gives the compact reading guide used throughout the paper. The metrics are intentionally simple: success is verifier-settled, value is pre-registered, and efficiency is value divided by spend. This makes the frontier interpretable through standard verified outcomes and a pre-registered value ledger.

Term	Reading
Batch-Level Task Budget Allocation	A task batch shares one hard budget; a 30-task run spends at most \$9.95 total.
Task Value	Pre-registered value gained when the task passes the Verifier.
Estimated Token Cost	Pre-run token-cost estimate used for cap planning.
Model Fit	Evidence about which model tier is likely to help this task.
Verifier	Domain outcome rule; SWE-bench uses a test-based harness.
Shared Budget	Hard batch-level resource constraint shared by all tasks.
Resolved Rate	Verifier-passed tasks divided by all tasks; 16/30 means 53.3%.
Total Resolved Value (TRV)	Sum of Task Value over verifier-passed tasks.
TRV per Dollar	TRV divided by total spend; TRV 18.0 at \$9.95 gives 1.81 TRV/\$.

Table 1. Core terms and metrics. The examples show how to read the reported numbers.

The Verifier is an outcome settlement rule. The policy uses Task Value, Estimated Token Cost, Model Fit, and Shared Budget state for allocation; the Verifier settles the final outcome after execution. In our SWE-bench instantiation, V_i is the test-based harness. This gives a strict external success signal while preserving the task-batch allocation structure.

The interfaces have different portability requirements. Task Value must be fixed before execution. Estimated Token Cost is a pre-run planning estimate; final spend is measured after execution. Model Fit can come from catalog priors, historical outcomes, or calibration runs. The Verifier must be a trusted outcome rule for the domain. The Shared Budget is set at the batch level. Their concrete calibration changes by domain, but the allocation problem consumes the same interface.

This interface also gives a domain-neutral governance rule. The policy allocates budget according to declared task value, estimated token cost, model-fit evidence, and the shared budget. In an enterprise batch, a coding task, a support task, and a document task can therefore compete under the same budget contract once each supplies the required signals and verifier.

4. BudgetFlow

BudgetFlow maintains a Shared Budget ledger across the task batch. For each task, it estimates whether the next model opportunity should use the cheap model, the strong model, or defer additional spend. The ledger is hard: Shared Budget gates each planned action. The goal is to turn the general interface in Section 3 into a concrete value-aware

Interface	SWE-bench instantiation	Other task-batch instantiations
Task Value	pre-registered criticality value	business priority, user tier, task owner value
Estimated Token Cost	pre-run token-cost estimate from task metadata	expected context size, tool steps, artifact size
Model Fit	tier evidence from prior runs and catalog priors	domain-specific success prior by model tier
Verifier	test-based harness	human review, rule check, delayed business outcome
Shared Budget	batch-level cap in dollars	team budget, project budget, processing window budget

Table 2. Domain interfaces for Batch-Level Task Budget Allocation. The paper evaluates the SWE-bench instantiation.

allocation policy.

The policy uses three allocation inputs:

- v_i , the Task Value;
- d_i , the Estimated Token Cost;
- $m_{i,k}$, the Model Fit for tier k .

These inputs enter distinct decisions. Task Value prioritizes tasks that create more value if verified. Estimated Token Cost becomes cap planning: the policy estimates how much token cost a task may need before it can produce a useful artifact. Model Fit estimates whether the strong model is likely to improve the outcome relative to the cheap model. Shared Budget controls affordability and stop/defer decisions.

Let h denote budget pressure, with higher h indicating tighter Shared Budget state. Let k_c be the cheap model and k_s be the strong model. A simplified task-start score for giving task i a strong-model opportunity is

$$S_i = \frac{v_i \cdot \max(m_{i,k_s} - m_{i,k_c}, 0)}{\max(d_i, \epsilon)} \cdot q(h),$$

where d_i is the task’s Estimated Token Cost and $q(h)$ reduces strong-model access as Shared Budget tightens. The implementation uses this score with affordability checks, learned task-router priors, and hard-budget gates. The formula captures the core allocation principle: strong-model budget is most valuable when a task has high value, sufficient expected model-fit gain, manageable Estimated Token Cost, and enough Shared Budget.

This makes BudgetFlow a governance layer over task categories. The score can favor a coding issue, a spreadsheet repair, or a support response for the same reason: the task has high declared value, a feasible token-cost estimate, useful model-fit upside, and enough shared budget to make the model opportunity affordable.

BudgetFlow separates allocation from outcome settlement. It uses Task Value, Estimated Token Cost, Model Fit, and Shared Budget state to choose model opportunities. The

Algorithm 1 BudgetFlow batch policy sketch

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1: Initialize Shared Budget state  $R \leftarrow B$ .
2: for task  $i$  in the batch do
3:   Read  $(v_i, d_i, m_{i,k_c}, m_{i,k_s})$  and current  $R$ .
4:   Set an initial task cap from Task Value, Estimated Token Cost, and  $R$ .
5:   Choose cheap or strong model from Model Fit, value upside, and affordability.
6:   Execute the task while updating spend and Shared Budget.
7:   Continue, stop, or defer using value upside and budget pressure.
8:   Settle the final artifact with  $V_i$  and record  $r_i$ .
9: end for
10: Report Resolved Rate, TRV, TRV per Dollar, and total spend.
    
```

Verifier evaluates the resulting patch or artifact after the task attempt. This design keeps the value objective auditable: the policy can explain why a task received budget, and the verifier independently determines whether that spend produced resolved value.

Allocation decisions. At task start, BudgetFlow assigns a route and an initial cap from value, Estimated Token Cost, Model Fit, and Shared Budget. During execution, it can assign a strong-model opportunity when the strong model has expected upside and Shared Budget allows it. It can stop or defer when additional spend has low expected value or would threaten the batch budget. These decisions are task-level in the current paper; finer stage or segment routing is left for future work.

Algorithm 1 is the implementation-facing view of the framework. The policy reads the three allocation interfaces before and during execution, while the Verifier is applied after the task attempt. This keeps routing, cap planning, and strong-model opportunity decisions auditable under the same value ledger used for evaluation.

5. Experimental Setup

Task set. We evaluate on fixed 30-task SWE-bench-style bug-fixing batches. Each task requires an agent to inspect a repository, edit code, and produce a patch. Outcomes are scored by the local harness and cross-checked with exported official predictions where available. Task values are pre-registered before execution.

Policies. Table 3 compares the five policies by the inputs they use. All policies run under the same shared hard budget and the same verifier. The key contrast is whether Task Value enters allocation of scarce model opportunities.

The 4-policy run uses the four policies available in that experimental line. All mainline policies use the same task order, budget protocol, model-tier catalog, and scoring path inside a run.

Metrics. The main tables report Resolved Rate, TRV, and TRV per Dollar, as defined in Table 1. Resolved Rate is written as both count and percentage. Spend is included as budget context because TRV per Dollar depends on it.

Sensitivity analyses. The main paid runs show observed frontier points. The sensitivity analyses ask how that frontier changes when the value ledger, the budget cap, or the serving cost model changes while the completed outcomes stay fixed. Task Value sensitivity re-scores fixed outcomes under alternate value profiles. Budget sensitivity replays the observed task order under different shared budget caps. KV Cache Cost-Discount sensitivity re-costs repeated input tokens under explicit cache-discount assumptions. Together, these diagnostics separate outcome quality from the accounting assumptions that determine cost-value efficiency.

6. Results

6.1. Main Paid Evidence

Table 4 summarizes four paid task-batch runs through the BudgetFlow frontier interface. Red entries mark BudgetFlow frontier signals; blue entries mark the strongest control when a control defines the observed frontier. The table should be read as a frontier-mode report: the same shared-budget objective can be shaped by Value Divergence, Strong Coverage, or Model Fit.

The **latest audited 5-policy run is the clearest positive policy result**: BudgetFlow resolves 16/30 tasks, reaches TRV 18.00, and achieves 1.81 TRV per Dollar. The next-best learned task-router resolves 15/30 tasks, reaches TRV 17.00, and achieves 1.71 TRV per Dollar. The 4-policy run gives a complementary value-governance result: BudgetFlow reaches TRV 18.50 and 2.32 TRV per Dollar while strong-model-only resolves one more task but reaches lower

value and lower value efficiency.

The runs show why BudgetFlow is useful beyond a single point comparison. First, value-aware allocation can protect high-value tasks and **improve TRV and TRV per Dollar** even when a strong-model policy resolves one additional task. Second, BudgetFlow exposes cases where **a strong model is already a strong frontier point** because broad strong-model coverage is cost-effective under the cap. Third, the latest audited policy shows the desired frontier movement: BudgetFlow **leads on standard resolution, value, and value per dollar** under the same cap.

These outcomes should be read together. The 4-policy run is the clearest Value Divergence case: the resolved tasks under BudgetFlow **carry more value and cost less**. The strong-model boundary runs show the Strong Coverage view: strong-model-only is valuable when the strong tier covers enough tasks under the shared cap. The router diagnostic shows the Model Fit view: a learned router is valuable when it places expensive model opportunities on tasks where the strong tier is productive. BudgetFlow is centered on the Value Divergence view while retaining these other views as diagnostic signals for future policy learning.

6.2. Frontier Mode Analysis

Table 5 summarizes the mechanism exposed by each main operating regime. This analysis is the interpretation layer for the same Resolved Rate, TRV, and TRV per Dollar evidence in Table 4. A shared-budget batch can be frontier-shaped by different signals: value divergence, broad strong-model coverage, or accurate model-fit routing. The BudgetFlow contribution is to make these signals explicit under one value-governed batch objective.

6.3. Full Latest 5-Policy Table

Table 6 gives the full latest audited run. **BudgetFlow leads all three primary metrics**. This is the clearest point-wise evidence for the current policy instantiation; the next section adds sensitivity curves to show how the same framework reports alternate operating conditions.

6.4. Task-Level Model Advantage

The latest run contains real model-tier heterogeneity. Among tasks where both cheap and strong model rows consumed model budget, there are **six cheap-model-favorable passes, five strong-model-favorable passes, and seven comparable failures under both**. This is the mechanism-level reason a batch-budget framework is useful: the task batch contains cheap-model opportunities, strong-model opportunities, and ceiling tasks in the same Shared Budget.

This task-level heterogeneity motivates selective strong-

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Policy	Same budget	Task Value	Token-cost cap planning	Model choice	Same verifier
cheap-model-only	yes			fixed cheap model	yes
strong-model-only	yes			fixed strong model	yes
learned task-router	yes			value-blind learned choice	yes
budget-only	yes		yes	budget-pressure choice	yes
BudgetFlow	yes	yes	yes	value-aware choice	yes

Table 3. Policy inputs and shared outcome contract. Token-cost cap planning is the decision-facing use of Estimated Token Cost.

Run	Leading policy	Resolved Rate	Spend	TRV	TRV per Dollar
4-policy value-aware allocation case	BudgetFlow	14/30 (46.7%)	\$7.98	18.50	2.32
	strong-model-only	15/30 (50.0%)	\$9.46	18.00	1.90
5-policy strong-model boundary case	strong-model-only	17/30 (56.7%)	\$9.95	20.00	2.01
	BudgetFlow	15/30 (50.0%)	\$9.95	17.50	1.76
5-policy learned-router stress case	strong-model-only	18/30 (60.0%)	\$9.88	19.00	1.92
	BudgetFlow	17/30 (56.7%)	\$9.95	18.50	1.86
Latest audited 5-policy positive case	BudgetFlow	16/30 (53.3%)	\$9.95	18.00	1.81
	learned task-router	15/30 (50.0%)	\$9.95	17.00	1.71
	strong-model-only	15/30 (50.0%)	\$9.95	16.50	1.66

Table 4. Main paid evidence. Each row reports the leader for one 30-task batch. Resolved Rate, TRV, and TRV per Dollar are the three primary metrics. Red bold entries mark BudgetFlow frontier signals; blue bold entries mark the strongest control in boundary runs. The 4-policy BudgetFlow row uses total observed spend, including non-scoreable abort overhead.

model use. When strong-model upside is concentrated on a subset of tasks, the allocation policy should preserve strong-model opportunities for those tasks while spending less on tasks where the cheap model is sufficient or both tiers are unlikely to pass.

7. Sensitivity Analysis

The sensitivity section has a different role from the main paid evidence. The main evidence reports observed runs; sensitivity analysis shows how the same completed outcomes move under alternate value, budget, and serving-cost assumptions. We therefore treat sensitivity as a frontier diagnostic: each table or curve explains how the operating view changes when the accounting environment changes.

7.1. Task Value Sensitivity

Task Value sensitivity tests the Value Divergence view under alternate ledgers. Table 7 re-scores the latest audited run under alternate Task Value profiles. BudgetFlow remains ahead of the best control under equal value, criticality value, compressed criticality, and expanded criticality. A value permutation diagnostic over 64 permutations gives BudgetFlow a positive margin in all samples, with median margin +1.00 TRV. The conclusion is that the latest frontier movement is visible across several value ledgers, including the plain resolved-count view where all resolved tasks have equal value.

7.2. Budget Sensitivity

Budget sensitivity is the closest analogue to a frontier plot. Figure 2 fixes observed outcomes and replays budget accounting at different caps. It therefore describes the observed cost-value frontier for this completed run. The strong-model-only policy is strongest at some low and mid caps, while BudgetFlow reaches the observed frontier at the full cap. The important reading is the frontier shape: budget pressure can move a batch between Strong Coverage, Model Fit, and Value Divergence regimes, and BudgetFlow provides the interface that makes this movement reportable.

7.3. KV Cache Cost-Discount Sensitivity

KV Cache Cost-Discount sensitivity tests a serving-cost question. Modern serving substrates can make repeated input context cheaper; this changes effective spend while leaving verified outcomes unchanged. Table 8 re-costs repeated input tokens under KV-cache discounts. BudgetFlow leads or stays close through KV90. At KV98 and KV99, the learned task-router has the highest TRV per Dollar, while BudgetFlow keeps a higher TRV than the learned task-router in the fixed-outcome table.

This pattern gives a useful serving-cost reading. BudgetFlow gains part of its efficiency by allocating scarce budget across tasks. When cache discounts become extreme, the cost penalty of repeated or stronger model use shrinks, and value-blind controls can become more competitive on TRV per Dollar. The result makes the serving boundary explicit: BudgetFlow is most useful under meaningful budget pres-

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Mode	Operating condition	Evidence	Interpretation
Value Divergence	Task values differ and model gains are selective.	In the 4-policy case, BudgetFlow resolves 14/30 but reaches TRV 18.50 and 2.32 TRV/\$, above strong-model-only's TRV 18.00 and 1.90 TRV/\$.	BudgetFlow is most valuable when the batch must choose which unequal-value tasks deserve scarce strong-model opportunity.
Strong Coverage	The strong model covers the batch broadly under the cap.	In the strong boundary case, strong-model-only reaches 17/30, TRV 20.00, and 2.01 TRV/\$, above BudgetFlow's 15/30, TRV 17.50, and 1.76 TRV/\$.	Broad strong-model coverage can form a frontier boundary that BudgetFlow should recognize.
Model Fit	The learned routing signal places model tiers accurately.	In the router diagnostic, learned routing reaches 19/30, TRV 21.00, and 2.11 TRV/\$, with 5/7 productive T3-starts versus 2/7 for BudgetFlow.	Accurate Model Fit is a reusable signal that should be absorbed into future value-aware policies.

Table 5. Frontier mode analysis. Each mode explains which signal limits or moves the observed cost-value frontier under the same batch-level objective.

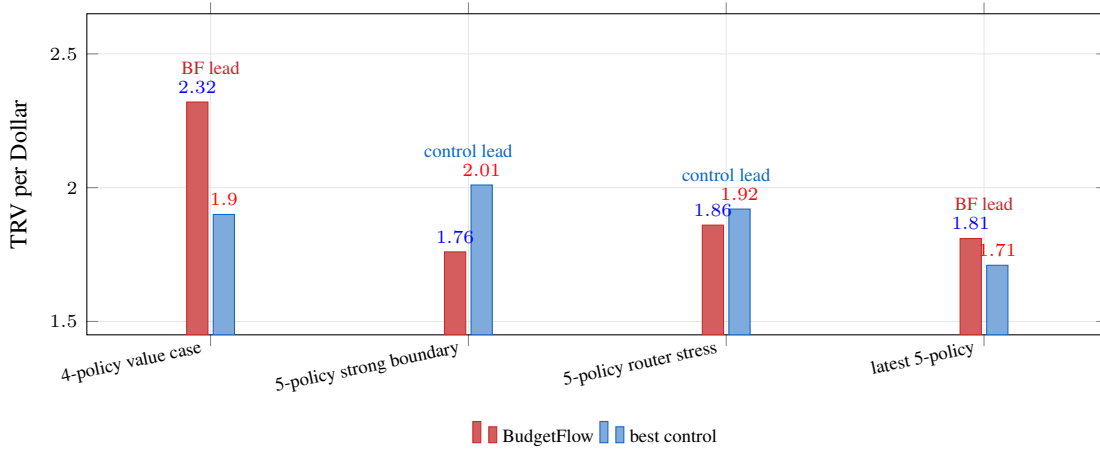


Figure 1. Multi-run main evidence. Each pair compares BudgetFlow with the best available control in that run on TRV per Dollar. The figure separates the repeated-run evidence from the sensitivity analysis in Section 7.

Policy	Resolved Rate	TRV	TRV/\$
BudgetFlow	16/30 (53.3%)	18.00	1.81
learned task-router	15/30 (50.0%)	17.00	1.71
strong-model-only	15/30 (50.0%)	16.50	1.66
budget-only	12/30 (40.0%)	13.00	1.31
cheap-model-only	11/30 (36.7%)	12.00	1.21

Table 6. Latest audited 5-policy run over 30 tasks.

Value profile	BF TRV	Control TRV	Margin
equal	16.00	15.00	+1.00
criticality	18.00	17.00	+1.00
compressed	17.00	16.00	+1.00
expanded	20.00	19.00	+1.00

Table 7. Task Value sensitivity on the latest audited run.

sure, and serving optimizations can move the batch toward Strong Coverage or Model Fit regimes.

8. Discussion and Future Work

Formulation-level generalization. The SWE-bench experiments instantiate the formulation in a verified coding domain. The formulation applies to task domains where tasks expose Task Value, Estimated Token Cost, Model Fit, and a Verifier. Coding tasks use tests as the Verifier.

Support tasks can use policy checks and service outcomes. Spreadsheet tasks can use cell-level checks or rule engines. Research and document-production tasks can use human review or domain checks. The calibration of each interface is domain-specific; the allocation objective and budget ledger are shared. This is the practical generalization claim: the task source changes, while the budget-governance contract remains the same.

Frontier-aware reporting. The same BudgetFlow framework explains multiple frontier positions across the paid

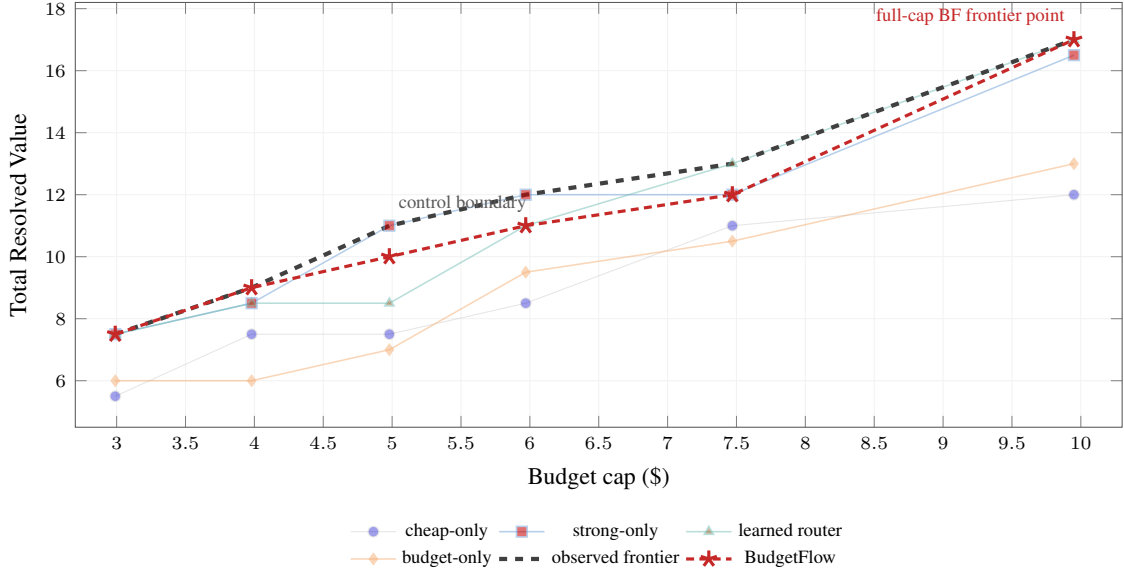


Figure 2. Budget sensitivity diagnostic on the latest audited run. Outcomes are fixed; budget accounting is replayed at alternate caps. The dashed line is the observed frontier; BudgetFlow is emphasized in red, while controls are shown as lighter diagnostic curves.

KV profile	BF	Best control	BF margin
KV0	1.81	1.71	+0.10
KV50	3.26	3.14	+0.11
KV90	9.06	8.98	+0.09
KV98	14.08	14.28	-0.20
KV99	15.13	15.41	-0.29

Table 8. KV Cache Cost-Discount sensitivity. Values are TRV per Dollar.

runs. Strong Coverage captures the common intuition that a broadly effective strong model can be worth its cost. Model Fit captures the learned-routing intuition that the best tier depends on the task. Value Divergence captures the BudgetFlow contribution: when tasks have unequal value and compete for one shared budget, the policy must decide where scarce strong-model opportunities create the most verified value. Reporting the frontier turns these cases into an operating guide for shared-budget agent batches.

Future work. Fine-grained task execution control. BudgetFlow can move from task-level allocation to finer-grained execution control inside a task. The current paper assigns model opportunities at the task level. A richer policy could decide which task stage or segment deserves the strong model, how much repair runway to preserve, and when stop/defer decisions should be made inside a long attempt. This direction should be evaluated against task-level controls because finer granularity can improve allocation only when the stage signal is reliable enough to justify the added switching cost.

Continual policy learning with auditability. Completed batches produce cost traces, routing outcomes, verifier results, and frontier positions. **A future policy can use those records to update future caps, route choices, and stop/continue decisions.** The important systems constraint is that each learned update should remain inspectable: the next allocation should be explainable through the same value ledger, Estimated Token Cost estimate, model-fit evidence, Shared Budget state, and verifier-settled outcome history.

Serving-aware and multi-tenant BudgetFlow. Serving systems such as vLLM, SGLang, and Dynamo optimize execution substrates: batching, cache locality, scheduling, and inference throughput. BudgetFlow operates at the task-budget layer above that substrate. A serving-aware BudgetFlow could pass task value, Shared Budget state, Estimated Token Cost, and outcome history into scheduling hints such as priority, continuation, model opportunity, and cache-sticky execution. The same interface can then extend from one budget owner to multiple teams, users, or services that share an agent execution substrate with distinct budgets and priorities. **This turns Batch-Level Task Budget Allocation into a candidate control plane for multi-tenant agent workloads.**

9. Conclusion

This paper formulates Batch-Level Task Budget Allocation for multi-step agent tasks under a shared hard budget. The core question is which task should receive the next scarce model opportunity when tasks have different value, different estimated token cost, and different model-tier fit. Budget-

Flow answers this question with a simple auditable interface: Task Value, Estimated Token Cost, Model Fit, Shared Budget state, and a Verifier-settled outcome.

The practical role of TRV is value accounting. It is a frozen value score that asks whether the same spend resolves more pre-declared task value. This makes mixed frontier positions meaningful: the paper reports which operating view shapes the cost-value frontier under a Shared Budget.

The SWE-bench experiments instantiate this interface in a verified coding domain. In the latest audited 5-policy run, BudgetFlow **improves Resolved Rate, TRV, and TRV per Dollar**. In the 4-policy value case, BudgetFlow resolves one fewer task than strong-model-only while producing higher TRV and higher TRV per Dollar. Across the broader paid evidence, the frontier is explained by three operating views. Strong Coverage describes batches where broad strong-model use is cost-effective under the cap. Model Fit describes batches where learned routing accurately places model tiers. Value Divergence describes the BudgetFlow regime: task values are uneven, model-tier advantage is selective, and the Shared Budget forces cross-task tradeoffs.

The broader value of BudgetFlow is the formulation. Coding tasks provide a strict verifier and measurable spend, but the same batch-budget structure appears whenever many agent tasks compete for one resource pool and produce outcomes with unequal value. By **separating allocation signals from outcome settlement**, BudgetFlow offers a portable way to reason about verified value under fixed resource constraints, from SWE-bench-style coding to future serving-aware and multi-tenant agent systems.

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